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Cite as: J. Chem. Phys. **117**, 593 (2002); <https://doi.org/10.1063/1.1480271>

Submitted: 14 January 2002 • Accepted: 02 April 2002 • Published Online: 24 June 2002

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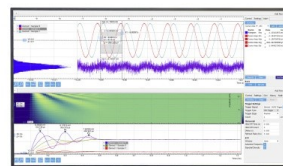
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Flexible transition state theory for a variable reaction coordinate: Derivation of canonical and microcanonical forms with angular momentum conservation

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(Received 14 January 2002; accepted 2 April 2002)

This paper extends work previously presented [J. Chem. Phys. **113**, 2648 (2000)] to establish a rigorous expression for the angular momentum resolved microcanonical reactive flux. A kinetic energy expression in terms of the angular momentum, based on the coordinate system described in the above reference, is derived. Analytic integration over momenta conjugate to the coordinates that describe the relative orientation of the fragments is effected at the canonical level. The Laplace transform relation between canonical and microcanonical coefficients is then exploited to yield the angular momentum dependent microcanonical reactive flux. Evaluation of the resulting rate expression involves a numerical integral whose dimension, in most cases, is the same as the integrals in corresponding canonical and standard microcanonical forms. © 2002 American Institute of Physics. [DOI: 10.1063/1.1480271]

I. INTRODUCTION

Despite many important advances in the application of molecular quantum dynamics to gas kinetics in the past two decades, statistical theories such as transition state theory¹ (TST) and the statistical adiabatic channel model^{2,3} (SACM) remain the only practical route by which gas phase rate coefficients can be estimated for reactions involving a large number of heavy atoms. As a consequence, the development of TST, and in particular the variant introduced by Wardlaw and Marcus,⁴ known as flexible transition state theory (FTST), is an active area of investigation. The principal motivation for the development of FTST was to extend the application of TST to reactions that exhibit large amplitude motion at the transition state (TS). An important class of reactions in this regard is recombination reactions that take place on a potential energy surface (PES) having no maximum along the reaction coordinate and thus no obvious TS location. Barrierless reactions are not confined to radical association/dissociation but include bimolecular reactants yielding a pair, or several pairs, of molecular products via a collision complex having a purely attractive potential in some or all arrangement channels, including the reactant channel. Consequently, the TS must be located variationally, and it is often the case that the location occurs at large values of the reaction coordinate (RC) where large amplitude motion, particularly rotation, is prevalent.

The variational procedure can be applied within a number of ensembles, the most common by far being the canonical and microcanonical, and variants thereof. The direct approach is to optimize the canonical rate coefficient, which has the advantage of minimizing the computational effort.

The rate coefficient is defined as a function of temperature T , and reaction coordinate R , i.e., $k = k(T, R)$. The value of k at $R^\ddagger$, where $R^\ddagger$ is defined by

$$\left(\frac{\partial k}{\partial R} \right)_{R=R^\ddagger} = 0, \quad (1)$$

is taken to be the optimum value of the canonical rate coefficient. This method will work well if the TS for all collision energies is located in approximately the same place, i.e., if there is a large well-defined barrier to reaction along the RC. For barrierless reactions this is clearly not the case, indeed, the location of the TS is a strong function of the overall energy of the combining fragments, in general moving to smaller values of R as the energy increases. For this case a more accurate estimate of the experimentally determined thermal rate coefficient is obtained by optimizing the microcanonical rate coefficients with respect to the RC and then forming the canonical coefficient by taking the appropriate canonical average. In terms of standard RRKM theory this amounts to optimizing the sum of states at the TS, $N(E, R)$, with respect to R , where E is the total energy.

Energy, of course, is not the only quantity that is conserved during an associative collision; angular momentum is also conserved. For barrierless association reactions this conservation has profound consequences, in that the centrifugal barrier is a major determinant of the location of the TS. Wardlaw and Marcus⁴ address this problem in their original formulation of FTST by determining the sum of states as a function of E , the magnitude of the total angular momentum, J , and the reaction coordinate RC, i.e., $N(E, J, R)$. The sum of states was then optimized with respect to R and the

canonical rate coefficient was then retrieved by integrating over J and forming the canonical average. It is important to recognize that prior to optimization

$$N(E, R) = \int dJ N(E, J, R). \quad (2)$$

It is the object of this paper to determine $N(E, J, R)$ and Eq. (2) will be used in this identification. It is worth noting that while Eq. (2) is true prior to optimization there is no similar expression postoptimization. In fact,

$$N(E; R_E^\ddagger) \geq \int dJ N(E, J; R_{EJ}^\ddagger). \quad (3)$$

This is because optimization of $N(E, R)$ effectively selects an average value for the RC, $R_E^\ddagger$, which does not necessarily correspond to the value $R_{EJ}^\ddagger$ minimizing $N(E, J, R)$ for a given (E, J) combination.

In their original formulation Wardlaw and Marcus⁴ expressed the necessary phase integral in terms of angle-action variables. In later work Smith recognized that considerable simplification, and consequently much insight, could be gained by using a more conventional set of coordinates. The specific simplification is that the kinetic energy term of the Hamiltonian is a quadratic form and so direct integration over the conjugate momenta is possible and halves the dimensionality of the phase integral. Smith⁵ developed expressions for the microcanonical rate coefficient (both dependent and independent of J) using a reaction coordinate that was based on the center-of-mass separation of the combining fragments.

The use of the center-of-mass separation as the reaction coordinate has a number of advantages in that it greatly simplifies the algebra required in the analytic integration over the conjugate momenta. For a number of important systems, e.g., the much studied $\text{CH}_3 + \text{CH}_3$ and the $\text{CH}_3 + \text{H}$ combinations, the center of mass (CM) reaction coordinate is similar in value to the bond reaction coordinate because the center-of-mass of the methyl fragment is located close to the carbon nucleus. Thus for the $\text{CH}_3 + \text{CH}_3$ and the $\text{CH}_3 + \text{H}$ systems, the choice of CM versus bond reaction coordinate, or any other reasonable choice of reaction coordinate, has little effect on the value of the optimized rate constant. However, Klippenstein recognized that in many, perhaps the vast majority of, reactions this will not necessarily be the case. An example is the association $^1\text{CH}_2 + \text{CO}$, where it is clear that the reaction coordinate is better approximated as the distance between the two bonding C atoms. Klippenstein⁶⁻⁸ developed methods that allow the optimization of the microcanonical rate coefficient to be performed with respect to a bond reaction coordinate and extensions to include general reaction coordinate optimization. In developing these methods Klippenstein also used direct analytic integration over the conjugate momenta for energy-resolved microcanonical rate coefficients that were independent of J . However, for energy and angular momentum resolved microcanonical rate coefficients Klippenstein encountered difficulties, in the coordinate system adopted for his approach, that prevented direct analytic integration and so resorted to numerical methods. Specifically, the associated energy and angular momentum resolved expressions contained singularities for

certain values of the angles describing the relative orientation of the fragments.

In a recent paper, here referred to as paper 3,⁹ the present authors reported an analysis of the reactive flux based on a different set of coordinates. This analysis applies to an arbitrary choice of the RC definition, the CM and bond coordinates being examples of two particular choices. The key to the approach was to use a body-fixed axis system and to describe the motion of the fragments relative to it. This approach has the advantage that the angles that describe the orientation of the body-fixed system are cyclic and can be integrated analytically. It was shown that analytic integration over the conjugate momenta was possible and the ensuing expressions for both canonical and microcanonical rate coefficients were given. The principal component of these expressions is a configuration integral (of dimension five or less) whose integrand contains the determinant of the inverse G-matrix associated with the external rotations and the relative internal rotation of the fragments. In paper 4,¹⁰ closed-form, analytic expressions were derived for the inverse G-matrix determinant for all combinations of fragment top types for an arbitrary reaction coordinate definition entirely in terms of kinetic energy matrix elements for a centers-of-mass reaction coordinate. In the present paper the same coordinate system will be used to obtain expressions for J -dependent canonical and microcanonical reaction fluxes.

The structure of the paper is as follows: In Sec. II a review of the pertinent material from paper 3 is presented with particular emphasis being given to the coordinate system used and the kinetic energy expressions based on this coordinate system. As demonstrated in paper 3 it is often easier to begin a derivation of a reactive flux at the canonical level as many of the integrals can be performed explicitly, and in Sec. III the canonical flux as a function of temperature and of angular momentum magnitude is derived. In Sec. IV the laplace transform relation between canonical and microcanonical ensembles is exploited, following the same sequence as paper 3, to obtain the angular momentum dependent microcanonical flux. Concluding remarks are gathered in Sec. V.

II. FLEXIBLE TRANSITION STATE THEORY

In general the canonical rate coefficient as a function of the reaction coordinate, R , can be written as^{11,12}

$$k(T, R) = \frac{g_e}{g_\sigma} \frac{1}{Q_{\text{react}}(T)} \frac{1}{h^N} \times \int d\mathbf{q} d\mathbf{p} \delta(f(\mathbf{q}) - R) \dot{f}(\mathbf{q}) u(\dot{f}) e^{-\beta H}. \quad (4)$$

Q_{react} is the partition function of the reactants, h is Planck's constant, and g_e and g_σ are the ratios of transition state to reactant electronic degeneracy factors and symmetry factors, respectively. In the phase space integral the vector $\mathbf{q}$ refers to the generalized coordinates and $\mathbf{p}$ to the momenta conjugate to $\mathbf{q}$, both being of dimension N . The function $f(\mathbf{q})$ defines the dividing surface (DS) separating reactants and products. The notation $\dot{f}$ is short hand for $\nabla f \cdot \dot{\mathbf{q}}$, the flux perpendicular to the dividing surface on the reaction coordinate. u is the

Heaviside unit step function. The overall Hamiltonian of the system is denoted H and $\beta = 1/k_B T$, where k_B is Boltzmann's constant.

As in paper 3 the approximation whereby degrees of freedom are classified as either conserved or transitional modes, first described by Wardlaw and Marcus, will be employed. In brief, conserved modes are those modes that are of low amplitude (and often high frequency) whose character changes only slightly on going from separated reactants to parent adduct. Transitional modes on the other hand tend to be large amplitude motions (e.g., free rotations) that become restricted (e.g., vibrations) as the parent adduct is formed and so change significantly in character. They also tend to be very strongly coupled to each other as well as to overall rotation and the RC, and for this reason overall rotational modes and the RC modes will be classified as transitional. However the two groups (conserved and transitional) are, to a good approximation, decoupled. This allows the overall Hamiltonian to be written as a sum, $H_c + H_{tr}$, H_c referring to the conserved modes and H_{tr} referring to transitional modes. Also, to a good approximation, the DS is generally only a function of the transitional coordinates, i.e., $f(\mathbf{q}) = f(\mathbf{q}_{tr})$. This allows the phase integral in Eq. (4) to be factored into two integrals: one over the conserved coordinates and the other over transitional coordinates. The integral over the conserved coordinates will be denoted Q_{cons} and is the canonical partition function for these modes. Standard expressions are available for Q_{cons} . With this factorization the rate constant expression becomes

$$k(T, R) = \frac{g_e}{g_\sigma} \frac{Q_{cons}(T)}{Q_{react}(T)} \frac{1}{h^{n+1}} \times \int d\mathbf{q}_{tr} d\mathbf{p}_{tr} \delta(R - f(\mathbf{q}_{tr})) \dot{f}(\mathbf{q}_{tr}) u(\dot{f}) e^{-\beta H_{tr}}, \quad (5)$$

where $n+1$ is the number of transitional/external rotational modes, including the RC. This expression can be written in a more compact form by identifying the canonical transitional mode flux $\mathcal{F}$ through the DS orthogonal to the reaction coordinate,

$$\mathcal{F}(T, R) = \frac{1}{h^{n+1}} \times \int d\mathbf{q}_{tr} d\mathbf{p}_{tr} \delta(R - f(\mathbf{q}_{tr})) \dot{f}(\mathbf{q}_{tr}) u(\dot{f}) e^{-\beta H_{tr}}. \quad (6)$$

Substituting this expression into Eq. (5) gives

$$k(T, R) = \frac{g_e}{g_\sigma} \frac{Q_{cons}(T, R) \mathcal{F}(T, R)}{Q_{react}(T)}. \quad (7)$$

The final rate coefficient is determined by invoking the variational principle and optimizing Eq. (7) with respect to variations in R .

Clearly, before any optimizing can be performed the integrals in Eq. (6) have to be evaluated. A crucial aspect of this evaluation is the selection of coordinate system. This

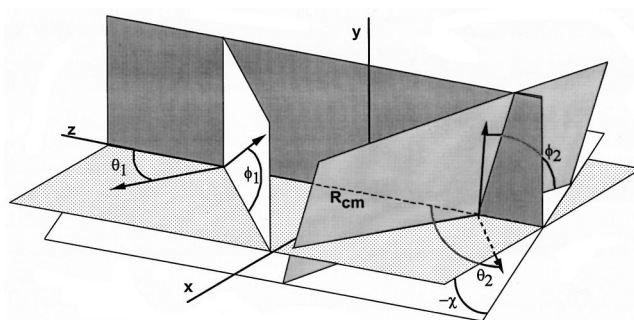


FIG. 1. The relative orientation of associating fragment principal axes with respect to the body fixed axes of the collision system.

selection was discussed at length in paper 3 and only a brief summary will be given here. To illustrate the coordinate system used in paper 3 this discussion will focus on the association of two asymmetric tops, since this represents the most complex (and in practice the most common) case of the five basic systems enumerated previously, the other cases being obtained from it by progressive simplification.

For two associating (effectively) rigid asymmetric tops, 12 degrees of freedom are required to specify the configuration of the system. Of these, three can be assigned to the overall center of mass of the system. The associated translational modes are decoupled from all other coordinates, can be safely factored out of the expression for the flux, and will not be discussed further. The remaining nine coordinates describe the transitional modes.

A key step in paper 3 was to define a body-fixed axis system such that the origin is at the overall center-of-mass of fragments and the z -axis lies along the vector between the centers-of-mass of the two fragments. (See Fig. 1 for a visualization of the body fixed system.) The y -axis of the body-fixed system is taken to be parallel with the line of nodes of one of the fragments (the choice of fragment is arbitrary and will be denoted fragment 1); the definition of the line of nodes will be discussed shortly. The direction of body-fixed x -axis follows automatically. Three Euler angles are then used to relate the orientation of the body-fixed axes to the space-fixed axes and these are denoted (Φ, Θ, Ψ) . This leaves six coordinates to assign. One of these will temporarily be assigned as the distance between the centers-of-mass of the two associating fragments and will be denoted R_{cm} . The remaining five coordinates are internal angles that describe the orientation of the fragments with respect to the *body-fixed* axes and, therefore, their relative orientation to each other. For each fragment a further set of internal axes can be defined; these are chosen to be the principal axes which have the property that the inertial tensor of each fragment is diagonal. In describing the orientation of the principal axes of each fragment with respect to the body-fixed axes of the overall system the rotational convention described by Zare¹³ will be adopted. In this convention each fragment is initially oriented such that its principal axes are aligned with the body-fixed axes. The final orientation of the fragment is then specified by three Euler angles: a rotation about the body-fixed z -axis by an angle ϕ_i (where i is either 1 or 2 depend-

ing on the fragment), followed by a rotation about an axis parallel to the body fixed y -axis (but passing through the center-of-mass of the fragment) by an angle θ_i and then, finally, a further rotation about the body-fixed z -axis by an angle χ_i . The rotation about the body-fixed y -axis of fragment 1 by θ_1 defines a line of nodes, i.e., the line of nodes of fragment 1 is the body-fixed y -axis. The consequence of this definition is that final rotation about the body-fixed z -axis of fragment 1 is coincident with the rotation of the body-fixed axes by the angle Φ about the space-fixed z -axis (using the standard y -convention), i.e., χ_1 for fragment 1 is Φ . This leaves just one χ angle, namely, χ_2 , which describes the dihedral angle between the two principal z -axes of the two fragments with respect to the body-fixed z -axes. (Henceforth the subscript will be dropped from χ_2 .) In summary, the generalized coordinates are described by the vector,

$$\mathbf{q}_{\text{cm}} = (\Phi, \Theta, \Psi, \phi_1, \theta_1, \phi_2, \theta_2, \chi, R_{\text{cm}}) \\ = (\Phi, \Theta, \Psi, \mathbf{q}_{\text{int}}, R_{\text{cm}}), \quad (8)$$

where the indices on the internal angles refer to fragment 1 and fragment 2. The subscript cm on $\mathbf{q}_{\text{cm}}$ indicates that the distance between fragments is determined by R_{cm} and replaces the tr subscript. The term $\mathbf{q}_{\text{int}}$ that appears in the second equality is for later notational convenience and is a vector of all the angles or internal coordinates that describe the relative orientation of the fragments.

The evaluation of Eq. (6) requires the expression of the integrand in the equation in terms of $\mathbf{q}_{\text{tr}}$ and $\mathbf{p}_{\text{tr}}$. For convenience this will first be done for the case of R_{cm} being the reaction coordinate. Then the expressions will be generalized for R being the reaction coordinate.

A. R_{cm} as reaction coordinate

Having defined the coordinate system that will be used the Hamiltonian can be written as

$$H_{\text{tr}} = T(\mathbf{p}_{\text{cm}}, \mathbf{q}_{\text{cm}}) + V_{\text{tr}}(\mathbf{q}_{\text{cm}}), \quad (9)$$

where $V_{\text{tr}}(\mathbf{q}_{\text{cm}})$ is the potential associated with the transitional modes. This potential will depend on the system in question and cannot be generalized. The term T denotes the kinetic energy of the transition modes and $\mathbf{p}_{\text{cm}}$ refers to the momentum conjugate to $\mathbf{q}_{\text{cm}}$. The general expression for the kinetic energy with reference to a body-fixed system is¹⁴

$$2T = \sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_i + 2\boldsymbol{\omega} \cdot \sum_{i=1}^N m_i \mathbf{r}_i \times \dot{\mathbf{r}}_i + \boldsymbol{\omega} \cdot \mathbf{I} \cdot \boldsymbol{\omega}, \quad (10)$$

where m_i is the mass of the atom i ; $\mathbf{r}_i = (x_i, y_i, z_i)$ is the position vector of atom i in the body-fixed frame; $\boldsymbol{\omega} = (\omega_x, \omega_y, \omega_z)$ is the instantaneous angular velocity relative to the body-fixed frame; and $\mathbf{I}$ is the instantaneous moment of inertia tensor for the entire system in the body-fixed frame. The angular momentum vector, $\mathbf{J}$, can be obtained from Eq. (10), in a formal sense, as,

$$\mathbf{J} = \frac{\partial T}{\partial \boldsymbol{\omega}} = \sum_{i=1}^N m_i \mathbf{r}_i \times \dot{\mathbf{r}}_i + \mathbf{I} \cdot \boldsymbol{\omega}. \quad (11)$$

As it stands the kinetic energy expression in Eq. (10) cannot be used to evaluate the reactive flux as it is not written in

terms of generalized coordinates (there is no vector of angles whose time derivative will give $\boldsymbol{\omega}$). The kinetic energy must first be expressed in terms of the generalized velocities associated with the transitional coordinates,

$$\dot{\mathbf{q}}_{\text{cm}} = (\dot{\Phi}, \dot{\Theta}, \dot{\Psi}, \dot{\phi}_1, \dot{\theta}_1, \dot{\phi}_2, \dot{\theta}_2, \dot{\chi}, \dot{R}_{\text{cm}}) \\ = (\dot{\Phi}, \dot{\Theta}, \dot{\Psi}, \dot{\mathbf{q}}_{\text{int}}, \dot{R}_{\text{cm}}). \quad (12)$$

However, the vector $(\dot{\Phi}, \dot{\Theta}, \dot{\Psi})$ is related to $\boldsymbol{\omega}$ by the well known similarity transform,

$$\mathbf{s} = \begin{bmatrix} -\sin \Theta \cos \Psi & \sin \Psi & 0 \\ \sin \Theta \sin \Psi & \cos \Psi & 0 \\ \cos \Theta & 0 & 1 \end{bmatrix} \quad (13)$$

(in the y -axis representation, see Goldstein¹⁵). Using this transformation an alternative velocity vector can be written as

$$\dot{\mathbf{Q}}_{\text{cm}} = (\boldsymbol{\omega}, \dot{\phi}_1, \dot{\theta}_1, \dot{\phi}_2, \dot{\theta}_2, \dot{\chi}, \dot{R}_{\text{cm}}) = (\boldsymbol{\omega}, \dot{\mathbf{q}}_{\text{int}}, \dot{R}_{\text{cm}}), \quad (14)$$

which is related to $\dot{\mathbf{q}}_{\text{cm}}$ by the linear transform,

$$\dot{\mathbf{Q}}_{\text{cm}} = \mathbf{S} \dot{\mathbf{q}}_{\text{cm}}, \quad (15)$$

where $\mathbf{S}$ is given by,

$$\mathbf{S} = \begin{pmatrix} \mathbf{s}_{3 \times 3} & \mathbf{0}_{3 \times m} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{m \times 3} & \mathbf{1}_{m \times m} & \mathbf{0}_{m \times 1} \\ \mathbf{0}_{1 \times 3} & \mathbf{0}_{1 \times m} & 1 \end{pmatrix}, \quad (16)$$

where m refers to the number of internal angles used to specify the relative orientation of the fragments, five in the current example, and $\mathbf{1}$ represents the unit or identity matrix.

The kinetic energy expression in Eq. (10) can be rewritten as a quadratic form in terms of $\dot{\mathbf{Q}}_{\text{cm}}$ as,

$$2T = \dot{\mathbf{Q}}_{\text{cm}}^T \cdot \mathbf{A}^0 \cdot \dot{\mathbf{Q}}_{\text{cm}}. \quad (17)$$

The 0 superscript will be used throughout to indicate a kinetic energy tensor that is based on the center-of-mass reaction coordinate. The structure of $\mathbf{A}^0$ is (see paper 3),

$$\mathbf{A}^0 = \begin{pmatrix} \mathbf{I}_{3 \times 3} & (\mathbf{A}_{\text{int-ext}}^0)_{3 \times m} & \mathbf{0}_{3 \times 1} \\ (\mathbf{A}_{\text{int-ext}}^0)^T_{m \times 3} & (\mathbf{A}_{\text{int}}^0)_{m \times m} & \mathbf{0}_{m \times 1} \\ \mathbf{0}_{1 \times 3} & \mathbf{0}_{1 \times m} & \mu \end{pmatrix}, \quad (18)$$

where $\mathbf{I}$, the instantaneous inertial tensor, and $\mathbf{A}_{\text{int}}^0$, kinetic energy matrix due to motion relative to the body-fixed system, are symmetric. The matrix $(\mathbf{A}_{\text{int-ext}}^0)_{3 \times m}$ represents Coriolis coupling between internal motion and overall rotation of the body-fixed frame. The element μ is the reduced mass of the associating fragments. Using the transformation given by Eq. (15) the kinetic energy in terms of the generalized velocities can be defined as

$$2T = \dot{\mathbf{q}}_{\text{cm}}^T \mathbf{S}^T \mathbf{A}^0 \mathbf{S} \dot{\mathbf{q}}_{\text{cm}} \quad (19)$$

$$= \dot{\mathbf{q}}_{\text{cm}}^T \mathbf{a}^0 \dot{\mathbf{q}}_{\text{cm}}. \quad (20)$$

The conjugate momentum vector $\mathbf{p}_{\text{cm}}$ can now be obtained by forming the Lagrangian and it is easy to show that

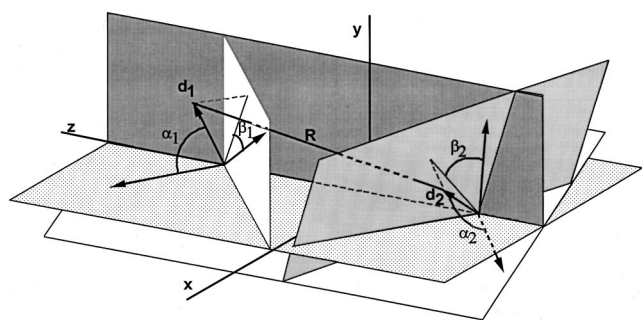


FIG. 2. The definition of the reaction coordinate distance R in terms of the vectors that define the pivot points relative to the centers-of-mass of each fragment ($\mathbf{d}_i$) and the R_{cm} distance.

$$\mathbf{p}_{\text{cm}} = \frac{\partial T}{\partial \dot{\mathbf{q}}_{\text{cm}}} = \mathbf{a}^0 \dot{\mathbf{q}}_{\text{cm}}. \quad (21)$$

Substitution of Eq. (21) into Eq. (20) results in an expression of T in terms of the integration variable $\mathbf{p}_{\text{cm}}$. In the case where R_{cm} is the RC, $f(\mathbf{q}_{\text{cm}})$ is in fact the last element in the $\mathbf{q}_{\text{cm}}$ vector and $\dot{f}(\mathbf{q}_{\text{cm}})$ is proportional to the last element in $\mathbf{p}_{\text{cm}}$. Thus the integrand in Eq. (6) is completely expressed in terms of the integration variables. (See paper 3 for a more detailed discussion.)

B. R as reaction coordinate

For a CM reaction coordinate $f(\mathbf{q}_{\text{cm}}) = R_{\text{cm}}$, the relative orientation of fragments along the RC is specified in Fig. 1 and the evaluation of Eq. (6), discussed at length in paper 3, is straightforward. As stated above, Klippenstein has shown the center-of-mass reaction coordinate is rarely the optimal reaction coordinate. A generalized description of the reaction coordinate requires a more general description of the separation of the fragments. That description is provided by Fig. 2, where R is the separation between a point in each fragment that is completely arbitrary. That point is specified by a vector $\mathbf{d}_i = (d_i, \alpha_i, \beta_i)$, in spherical coordinates, for the i th fragment and locates the point with respect to the internal principal axes of each fragment. A variable reaction coordinate form of FTST allows for optimization of both $\mathbf{d}_i$ and R in minimizing reactive flux as a function of total energy and angular momentum. Thus, in general $f(\mathbf{q}_{\text{cm}})$ is a more complex function of the components of $\mathbf{q}_{\text{cm}}$ and more work is required, the details of which are also given in paper 3. In brief, a new coordinate for measuring the distance between fragments is defined as

$$q_{n+1} = f(\mathbf{q}_{\text{cm}}), \quad (22)$$

and a coordinate transformation of the form,

$$\begin{aligned} \mathbf{q}_{\text{cm}} &= (\Phi, \Theta, \Psi, \mathbf{q}_{\text{int}}, R_{\text{cm}}) \rightarrow \\ \mathbf{q} &= (\Phi, \Theta, \Psi, \mathbf{q}_{\text{int}}, q_{n+1}) \end{aligned} \quad (23)$$

is made. In effect what this transform implies is that $f(\mathbf{q}_{\text{cm}})$ [Eq. (22)] can be inverted so that R_{cm} can be written as a function of all of the angles and q_{n+1} ,

$$R_{\text{cm}} = F(\mathbf{q}). \quad (24)$$

The coordinate transformation implies a relationship between the generalized velocities of the form,

$$\dot{\mathbf{q}}_{\text{cm}} = \mathcal{J}_q \dot{\mathbf{q}}, \quad (25)$$

where the Jacobean for this transform is,

$$\mathcal{J}_q = \begin{pmatrix} \mathbf{1}_{3 \times 3} & \mathbf{0}_{3 \times m} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{m \times 3} & \mathbf{1}_{m \times m} & \mathbf{0}_{m \times 1} \\ \mathbf{0}_{1 \times 3} & \mathbf{b}_{1 \times m} & \frac{\partial F}{\partial q_{n+1}} \end{pmatrix}, \quad (26)$$

where

$$\mathbf{b}_{1 \times m} = \left(\frac{\partial F}{\partial \phi_1}, \dots, \frac{\partial F}{\partial \chi} \right) \equiv \nabla_{\mathbf{q}_{\text{int}}} F. \quad (27)$$

Implicit in Eq. (26) is the fact that neither R_{cm} or q_{n+1} is dependent on the angles (Φ, Θ, Ψ) that orient the body-fixed system, i.e., R_{cm} can be written as $R_{\text{cm}} = F(\mathbf{q}_{\text{int}}, q_{n+1})$, the magnitude of these distances remaining invariant to an overall rotation of the system. The notation in Eq. (27) reflects this in that $\mathbf{q}_{\text{int}}$ refers only to the internal angles that define the relative orientation of the two fragments. An examination of the block diagonal structure of the transform matrices $\mathbf{S}$ and $\mathcal{J}_q$ shows that they commute, from which it follows through Eq. (19) that

$$2T = \dot{\mathbf{q}}^T \mathcal{J}_q^T \mathbf{S}^T \mathbf{A}^0 \mathbf{S} \mathcal{J}_q \dot{\mathbf{q}} \quad (28)$$

$$= \dot{\mathbf{q}}^T \mathbf{S}^T \mathcal{J}_q^T \mathbf{A}^0 \mathcal{J}_q \mathbf{S} \dot{\mathbf{q}} \quad (29)$$

$$= \dot{\mathbf{q}}^T \mathbf{S}^T \mathbf{A} \mathbf{S} \dot{\mathbf{q}} \quad (30)$$

$$= \dot{\mathbf{q}}^T \mathbf{a} \dot{\mathbf{q}}. \quad (31)$$

The form of the kinetic energy given by Eq. (30) will be particularly useful. Defining the vector $\dot{\mathbf{Q}}$ such that,

$$\dot{\mathbf{Q}} = (\boldsymbol{\omega}, \dot{\mathbf{q}}_{\text{int}}, \dot{q}_{n+1}), \quad (32)$$

it follows that $\dot{\mathbf{Q}} = \mathbf{S} \dot{\mathbf{q}}$ and so the kinetic equation can be written in yet another form, i.e.,

$$2T = \dot{\mathbf{Q}}^T \mathbf{A} \dot{\mathbf{Q}}, \quad (33)$$

which is of the same form as Eq. (10) when expanded using Eq. (32). The formal differentiation of Eq. (33) with respect to $\dot{\mathbf{Q}}$ yields a momentum related vector which will be denoted $\mathbf{p}_J$,

$$\mathbf{p}_J = \mathbf{A} \dot{\mathbf{Q}} = \mathbf{A} \mathbf{S} \dot{\mathbf{q}}. \quad (34)$$

Comparison with Eq. (11) shows that the first three elements of $\mathbf{p}_J$ are the elements of the total angular momentum $\mathbf{J}$, i.e., $\mathbf{p}_J = (\mathbf{J}, \mathbf{p}_{\text{int}}, p_{n+1})$, where $\mathbf{p}_{\text{int}}$ represents the momenta conjugate to $\mathbf{q}_{\text{int}}$ and p_{n+1} is the momentum along the RC.

The momentum $\mathbf{p}$ conjugate to $\mathbf{q}$ is obtained in the standard way via the Lagrangian and gives

$$\mathbf{p} = \mathbf{S}^T \mathbf{A} \mathbf{S} \dot{\mathbf{q}}. \quad (35)$$

Comparison of Eqs. (34) and (35) gives

$$\mathbf{p} = \mathbf{S}^T \mathbf{p}_J. \quad (36)$$

Also, it was shown in paper 3 that $\mathbf{p}$ and $\mathbf{p}_{\text{cm}}$ are related by

$$\mathbf{p}_{\text{cm}} = (\mathcal{J}_q^T)^{-1} \mathbf{p}, \quad (37)$$

and, as a consequence of this, the transform was shown to be canonical, i.e., $d\mathbf{q}_{\text{cm}}d\mathbf{p}_{\text{cm}} = d\mathbf{q}d\mathbf{p}$. The transitional mode flux [Eq. (6)] can be rewritten as

$$\mathcal{F}(\beta, R) = \frac{1}{h^{(n+1)}} \times \int d\mathbf{q}d\mathbf{p} \delta(q_{n+1} - R) \dot{q}_{n+1} u(\dot{q}_{n+1}) e^{-\beta H_{\text{tr}}}. \quad (38)$$

For later convenience this expression will be rewritten as

$$\mathcal{F}(\beta, R) = \frac{1}{h^{(n+1)}} \int d\mathbf{q} \delta(q_{n+1} - R) e^{-\beta V_{\text{tr}}} \mathcal{P}, \quad (39)$$

where the momentum integral is

$$\mathcal{P} = \int d\mathbf{p} \dot{q}_{n+1} u(\dot{q}_{n+1}) e^{-\beta T}, \quad (40)$$

the kinetic energy term is

$$2T = \mathbf{p}^T \mathbf{g} \mathbf{p}, \quad (41)$$

and $\mathbf{g}$ is defined such that $\mathbf{a}\mathbf{g} = \mathbf{1}$, $\mathbf{a}$ being defined by Eqs. (30) and (31).

At this point the necessary coordinate representation has been established to allow a partial integration over the momenta so as to obtain the canonical transitional mode flux as a function of R , β , and J .

III. EVALUATION OF THE CANONICAL FLUX

The first step in evaluating the canonical flux is to integrate over the momentum conjugate to the reaction coordinate, q_{n+1} . This was discussed in paper 3 and yields

$$\mathcal{P} = \beta^{-1} \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} d\mathbf{p}_r \exp\left(-\frac{\beta}{2} (\mathbf{p}_r^T \mathbf{g}_r \mathbf{p}_r)\right), \quad (42)$$

where the matrix $\mathbf{g}_r$ is the inverse of $\mathbf{a}_r$, the latter matrix being obtained from the matrix $\mathbf{a}$ by striking out the row and column associated with the reaction coordinate q_{n+1} . The vector $\mathbf{p}_r$ is likewise related to $\mathbf{p}$ by the removal of the element that is the momentum conjugate to q_{n+1} .

Now, from Eq. (16) of paper 3 we can express the integral in Eq. (42) in terms of the components of the angular momentum, $\mathbf{J}$, by using the transform $\mathbf{S}_r$ which is related to the transform $\mathbf{S}$ introduced in Sec. II by removing the row and column relating to the RC. With this transform it is easy to show that

$$\mathbf{a}_r = \mathbf{S}_r^T \mathbf{A}_r \mathbf{S}_r, \quad (43)$$

where $\mathbf{A}_r$ is similarly obtained from $\mathbf{A}$ by the elimination of the row and column related to the reaction coordinate.

Substitution of Eq. (43) into Eq. (42) gives

$$\mathcal{P} = \beta^{-1} \int d\mathbf{p}_r \exp\left(-\frac{\beta}{2} (\mathbf{p}_r^T \mathbf{S}_r^{-1} \mathbf{A}_r^{-1} (\mathbf{S}_r^T)^{-1} \mathbf{p}_r)\right). \quad (44)$$

By an identical set of arguments that was used to derive Eq. (36) it can be shown that

$$(\mathbf{S}_r^T)^{-1} \mathbf{p}_r = \mathbf{p}_{J,r}. \quad (45)$$

Again, $\mathbf{p}_{J,r}$ is derived from $\mathbf{p}_J$ by removing the element associated with the RC.

It follows from Eq. (45) that the Jacobean for the transform $\mathbf{p}_r \rightarrow \mathbf{p}_{J,r}$ is $\sin \Theta$. Applying this transformation to Eq. (44) gives

$$\mathcal{P} = \frac{\sin \Theta}{\beta} \int d\mathbf{p}_{J,r} \exp\left(-\frac{\beta}{2} (\mathbf{p}_{J,r}^T \mathbf{G}_r \mathbf{p}_{J,r})\right), \quad (46)$$

where $\mathbf{G}_r$ is the inverse of $\mathbf{A}_r$ (i.e., $\mathbf{A}_r \mathbf{G}_r = \mathbf{1}$).

From Eqs. (18) and (30) it can be shown that $\mathbf{A}_r$ has the following structure (see paper 3):

$$\mathbf{A}_r = \begin{pmatrix} \mathbf{I}_{3 \times 3} & (\mathbf{A}_{\text{int-ext}}^0)_{3 \times m} \\ (\mathbf{A}_{\text{int-ext}}^0)^T_{m \times 3} & (\mathbf{A}_{\text{int}}^0)_{m \times m} + \mu(\mathbf{b}\mathbf{b}^T)_{m \times m} \end{pmatrix}, \quad (47)$$

where the vector $\mathbf{b}$ was defined in Eq. (27). For ease of notation Eq. (47) will be written as

$$\mathbf{A}_r = \begin{pmatrix} \mathbf{B} & \mathbf{C} \\ \mathbf{C}^T & \mathbf{D} \end{pmatrix}. \quad (48)$$

Since $\mathbf{A}_r$ is symmetric, $\mathbf{G}_r$ is also symmetric and has a similar structure which for notational convenience will be written as

$$\mathbf{G}_r = \begin{pmatrix} \mathbf{W} & \mathbf{X} \\ \mathbf{X}^T & \mathbf{Z} \end{pmatrix}. \quad (49)$$

Substitution of Eq. (49) into Eq. (46) followed by expansion yields

$$\mathcal{P} = \frac{\sin \Theta}{\beta} \int d\mathbf{J} \exp\left(-\frac{\beta}{2} (\mathbf{J}^T \mathbf{W} \mathbf{J})\right) \times \int d\mathbf{p}_{\text{int}} \exp\left(-\frac{\beta}{2} (2\mathbf{J}^T \mathbf{X} \mathbf{p}_{\text{int}} + \mathbf{p}_{\text{int}}^T \mathbf{Z} \mathbf{p}_{\text{int}})\right). \quad (50)$$

The integral over $\mathbf{p}_{\text{int}}$ can be effected by completing the square in the exponent; let $\mathbf{c}$ be an arbitrary vector of the same dimension as $\mathbf{p}_{\text{int}}$, such that the quadratic form, $(\mathbf{p}_{\text{int}} + \mathbf{c})^T \mathbf{Z} (\mathbf{p}_{\text{int}} + \mathbf{c})$, can be defined. Thus the square in the exponent of Eq. (50) can be completed if,

$$\mathbf{c}^T \mathbf{Z} = \mathbf{J}^T \mathbf{X} \text{ or } \mathbf{c} = \mathbf{Z}^{-1} \mathbf{X}^T \mathbf{J}, \quad (51)$$

where the symmetry of $\mathbf{Z}$ has been invoked. Equation (50) can be rewritten as

$$\begin{aligned} \mathcal{P} &= \frac{\sin \Theta}{\beta} \int d\mathbf{J} \exp\left(-\frac{\beta}{2} (\mathbf{J}^T \mathbf{W} \mathbf{J})\right) \\ &\times \int d\mathbf{p}_{\text{int}} \exp\left(-\frac{\beta}{2} ((\mathbf{p}_{\text{int}} + \mathbf{c})^T \mathbf{Z} (\mathbf{p}_{\text{int}} + \mathbf{c}) - \mathbf{c}^T \mathbf{Z} \mathbf{c})\right) \\ &= \frac{\sin \Theta}{\beta} \int d\mathbf{J} \exp\left(-\frac{\beta}{2} (\mathbf{J}^T \mathbf{W} \mathbf{J} - \mathbf{c}^T \mathbf{Z} \mathbf{c})\right) \\ &\times \int d\mathbf{p}_{\text{int}} \exp\left(-\frac{\beta}{2} ((\mathbf{p}_{\text{int}} + \mathbf{c})^T \mathbf{Z} (\mathbf{p}_{\text{int}} + \mathbf{c}))\right). \end{aligned} \quad (52)$$

For the integral over $\mathbf{p}_{\text{int}}$, the vector $\mathbf{c}$ is constant thus the Jacobean for the transform $\mathbf{p}_{\text{int}} + \mathbf{c} \rightarrow \mathbf{p}'$ is unity, giving,

$$\mathcal{P} = \frac{\sin \Theta}{\beta} \int d\mathbf{J} \exp\left(-\frac{\beta}{2}(\mathbf{J}^T \mathbf{W} \mathbf{J} - \mathbf{c}^T \mathbf{Z} \mathbf{c})\right) \times \int d\mathbf{p}' \exp\left(-\frac{\beta}{2}((\mathbf{p}')^T \mathbf{Z} \mathbf{p}')\right). \quad (53)$$

The integral over $\mathbf{p}'$ is now of standard form and can be integrated directly to give

$$\mathcal{P} = \frac{\sin \Theta}{\beta} \int d\mathbf{J} \exp\left(-\frac{\beta}{2}(\mathbf{J}^T \mathbf{W} \mathbf{J} - \mathbf{c}^T \mathbf{Z} \mathbf{c})\right) \times \left(\frac{2\pi}{\beta}\right)^{(n-3)/2} \frac{1}{|\mathbf{Z}|^{1/2}}, \quad (54)$$

where $|\mathbf{Z}|$ represents the determinant of the matrix $\mathbf{Z}$. The exponent in the integral over $\mathbf{J}$ when expanded using Eq. (51) gives,

$$\mathbf{J}^T \mathbf{W} \mathbf{J} - \mathbf{c}^T \mathbf{Z} \mathbf{c} = \mathbf{J}^T (\mathbf{W} - \mathbf{X} \mathbf{Z}^{-1} \mathbf{X}^T) \mathbf{J} = \mathbf{J}^T \mathbf{B}^{-1} \mathbf{J}, \quad (55)$$

where Eq. (A10) has been applied in the second equality. The matrix $\mathbf{B}$ is the instantaneous inertial tensor matrix of the overall system. Thus Eq. (54) becomes

$$\mathcal{P} = \frac{\sin \Theta}{\beta} \left(\frac{2\pi}{\beta}\right)^{(n-3)/2} \frac{1}{|\mathbf{Z}|^{1/2}} \times \int d\mathbf{J} \exp\left(-\frac{\beta}{2}(\mathbf{J}^T \mathbf{B}^{-1} \mathbf{J})\right), \quad (56)$$

where the determinant $|\mathbf{Z}|^{-1/2} = (|\mathbf{Z}^{-1}|)^{1/2}$ can be written in terms of the determinants of the submatrices of $\mathbf{A}_r$ by applying Eq. (A12) from Appendix A. Explicit expressions for $\mathbf{A}_r$ are provided in the Appendix of paper 4.

The angular momentum components in Eq. (56) are defined with respect to the body-fixed axes so, without loss of generality, the integral Eq. (56) can be simplified by making an orthogonal transformation $\mathbf{J} \rightarrow \mathbf{J}'$ such that $\mathbf{B}^{-1}$ is diagonal giving,

$$\mathcal{P} = \frac{\sin \Theta}{\beta} \left(\frac{2\pi}{\beta}\right)^{(n-3)/2} \left(\frac{|\mathbf{A}_r|}{|\mathbf{B}|}\right)^{1/2} \times \int d\mathbf{J}' \exp\left(-\frac{\beta}{2}(J_x'^2/I_a + J_y'^2/I_b + J_z'^2/I_c)\right), \quad (57)$$

where I_a , I_b , and I_c are the instantaneous principal moments of inertia of the overall system and correspond to the eigenvalues of $\mathbf{B}$; these principal moments depend on $\mathbf{q}_{\text{int}}$. In Eq. (57) the $|\mathbf{Z}|^{-1/2}$ factor has been replaced by $(|\mathbf{A}_r|/|\mathbf{B}|)^{1/2}$. This replacement is derived from the fact that if the integration in Eq. (57) is carried out over the full $(-\infty, \infty)$ range of J_x' , J_y' , and J_z' then the result must be identical to Eq. (64) in paper 3.

The object of this paper is to obtain results in terms of the angular momentum magnitude, J . To realize this goal a further transform of the angular momentum components from cartesian to spherical polar coordinates is required which gives

$$\mathcal{P} = \frac{\sin \Theta}{\beta} \left(\frac{2\pi}{\beta}\right)^{(n-3)/2} \left(\frac{|\mathbf{A}_r|}{|\mathbf{B}|}\right)^{1/2} \times \int dJ d\eta d\nu J^2 \sin \eta \exp(-\beta E_{\text{rot}}(J)), \quad (58)$$

where

$$E_{\text{rot}}(J) = J^2([\sin^2 \nu/I_a + \cos^2 \nu/I_b] \sin^2 \eta + \cos^2 \eta/I_c)/2. \quad (59)$$

The overall flux can now be recovered by inserting Eq. (58) into Eq. (39). As noted in paper 3, in the absence of any external fields the transitional potential is independent of the three Euler angles, (Θ, Ψ, Φ) that describe the orientation of the body-fixed axes. Thus the integrand is independent of these angles so that integration over them can be done analytically. Also, the simple delta function in Eq. (39) allows analytic integration over q_{n+1} to give,

$$\mathcal{F}(\beta, R) = \frac{4\pi}{h^{(n+1)}} \left(\frac{2\pi}{\beta}\right)^{(n-1)/2} \int d\mathbf{q}_{\text{int}} \left(\frac{|\mathbf{A}_r|}{|\mathbf{B}|}\right)^{1/2} \times \int dJ d\eta d\nu J^2 \sin \eta \times \exp(-\beta(V_{\text{tr}} + E_{\text{rot}}(J))), \quad (60)$$

where Eq. (24) can now be expressed as

$$R_{\text{cm}} = F(\mathbf{q}_{\text{int}}, R). \quad (61)$$

(Contrast Figs. 1 and 2 for a visualization of the connection between R_{cm} and a completely general R .)

A J -dependent canonical flux, $\mathcal{F}(\beta, J, R)$, can be defined in an analogous way to that discussed for the microcanonical ensemble in the Introduction, and is related to the overall canonical flux in a similar way,

$$\mathcal{F}(\beta, R) = \int dJ \mathcal{F}(\beta, J, R). \quad (62)$$

By rearranging the order of integration in Eq. (60) and comparing with Eq. (62) the J -dependent canonical flux, is given by

$$\mathcal{F}(\beta, J, R) = J^2 \frac{4\pi}{h^{(n+1)}} \left(\frac{2\pi}{\beta}\right)^{(n-1)/2} \int d\mathbf{q}_{\text{int}} \left(\frac{|\mathbf{A}_r|}{|\mathbf{B}|}\right)^{1/2} \times \int_0^\pi d\eta \sin \eta \int_0^{2\pi} d\nu \times \exp(-\beta(V_{\text{tr}} + E_{\text{rot}}(J))). \quad (63)$$

Equation (63) forms the basis from which the microcanonical rate coefficient will be derived in the next section.

IV. MICROCANONICAL FTST FOR VARIABLE REACTION COORDINATE

Prior to reaction coordinate optimization the J -dependent microcanonical FTST rate constant expression is

$$k(E, J, R) = \frac{g_e}{g_\sigma} \frac{N(E, J, R)}{h \rho_{\text{react}}(E, J)}, \quad (64)$$

where g_σ and g_e are the same as in the corresponding canonical expression [Eq. (7)], $N(E, J, R)$ is the transition state sum of states (number of states on the microcanonical DS with energy less than or equal to E and angular momentum magnitude equal to J), and $\rho_{\text{react}}(E, J)$ is the density of states of the reactants. The canonical rate coefficient is obtained from $k(E, J, R)$ as

$$k(\beta, R) = \frac{g_e}{g_\sigma} \frac{1}{Q_{\text{react}}(\beta)} \times \int dE dJ k(E, J, R) \rho_{\text{react}}(E, J) e^{-\beta E} \quad (65)$$

$$= \frac{g_e}{g_\sigma} \frac{1}{h Q_{\text{react}}(\beta)} \int dE dJ N(E, J, R) e^{-\beta E} \quad (66)$$

$$= \frac{g_e}{g_\sigma} \frac{1}{h Q_{\text{react}}(\beta)} \int dJ \mathcal{L}[N(E, J, R)], \quad (67)$$

where $\mathcal{L}[\cdot]$ indicates a Laplace transform.

In terms of the conserved and transitional modes the numerator of Eq. (64) can be written as the convolution

$$h^{-1} N(E, J, R) = \int_0^E d\epsilon \rho_{\text{cons}}(\epsilon, R) h^{-1} N_{tr}(E - \epsilon, J, R) \\ = \int_0^E d\epsilon \rho_{\text{cons}}(\epsilon, R) f(E - \epsilon, J, R). \quad (68)$$

Here $\rho_{\text{cons}}(\epsilon, R)$ is the conserved mode density of states at energy ϵ and $f(\epsilon_{tr}, J, R)$ is the microcanonical transitional mode flux at energy $\epsilon_{tr} = E - \epsilon$ through a DS orthogonal to the RC. Note that both quantities depend on the value R of the RC, but that only f is regarded as a function of J , or to put it another way the conserved modes do not contribute significantly to the angular momentum, an approximation that appears to be universally adopted. $\rho_{\text{cons}}(\epsilon, R)$ is usually calculated on a quantum mechanical basis using direct count algorithms and will not be discussed further.

Insertion of Eq. (68) into Eq. (67) followed by the application of the convolution theorem gives,

$$k(\beta, R) = \frac{g_e}{g_\sigma} \frac{Q_{\text{cons}}}{Q_{\text{react}}(\beta)} \int dJ \mathcal{L}[f(E, J, R)]. \quad (69)$$

Comparison with Eq. (7) indicates that

$$\mathcal{F}(\beta, R) = \int dJ \mathcal{L}[f(E, J, R)], \quad (70)$$

or, by using Eq. (62),

$$\mathcal{F}(\beta, J, R) = \mathcal{L}[f(E, J, R)]. \quad (71)$$

Hence the inverse Laplace transform of $\mathcal{F}(\beta, J, R)$ will give the J -dependent microcanonical flux. Inversion of Eq. (63) gives

$$f(E, J, R) = \frac{2J^2}{\Gamma((n-1)/2)} \left(\frac{2\pi}{h^2} \right)^{(n+1)/2} \\ \times \int d\mathbf{q}_{\text{int}} \left(\frac{|\mathbf{A}_r|}{|\mathbf{B}|} \right)^{1/2} \int_0^\pi d\eta \sin \eta \int_0^{2\pi} d\nu \\ \times (E - [V_{tr} + E_{\text{rot}}(J)])^{(n-3)/2} \\ \times u(E - [V_{tr} + E_{\text{rot}}(J)]). \quad (72)$$

Equations (63) and (72) represent the main results of this paper. It is shown in Appendix B that it is possible to integrate analytically over η and, in most cases, ν , the final expressions being expressed in terms of elliptic integrals. Appendix B also describes approximations to the integral over ν that can be used when analytic expressions are not available. The remaining integrals are then solely in terms of the internal coordinates, $\mathbf{q}_{\text{int}}$, and, other than particularly simple cases, must be performed numerically, either by quadrature or using a Monte Carlo method.

V. SUMMARY

This paper has extended the results in paper 3 to include the effects of the conservation of angular momentum. The classical, J -dependent transitional mode configuration integral has been reduced to at most a six-dimensional integral which is just one dimension higher than the corresponding expressions in which the total angular momentum is not resolved but implicitly summed over. The same set of body-fixed coordinates are used, allowing the results obtained in paper 4 to be fully exploited. The final microcanonical expression for $k(E, J)$, after optimizing with respect to R , is,

$$k(E, J) = \frac{g_e}{g_\sigma} \frac{2J^2}{\Gamma((n-1)/2) \rho_{\text{react}}(E, J)} \left(\frac{2\pi}{h^2} \right)^{(n+1)/2} \\ \times \int_0^E d\epsilon \rho_{\text{cons}}(\epsilon, R^\dagger) \int d\mathbf{q}_{\text{int}} \left(\frac{|\mathbf{A}_r|}{|\mathbf{B}|} \right)^{1/2} \\ \times \int_0^\pi d\eta \sin \eta \int_0^{2\pi} d\nu (E - \epsilon - V_{\text{eff}}(R^\dagger))^{(n-3)/2} \\ \times u(E - \epsilon - V_{\text{eff}}(R^\dagger)), \quad (73)$$

where $R^\dagger$ is the optimal value of R for the choice of E and J and

$$V_{\text{eff}}(R) = V_{tr}(\mathbf{q}_{\text{int}}, R) + E_{\text{rot}}(J, \mathbf{q}_{\text{int}}, R). \quad (74)$$

V_{eff} represents the effective potential on which the reaction takes place for a given value of J , including as it does, terms arising from centrifugal effects, and is equivalent to the effective potential that occurs in scattering theory. The form of the determinant $|\mathbf{A}_r|$ has been discussed extensively in paper 3, and analytic expressions for the elements of $\mathbf{B}$ can be found in paper 4. As shown in Appendix B, the integration over η , and in most cases ν , can be done analytically.

The microcanonical rate expression (74) is broadly similar to that obtained recently by Smith⁵ but the coordinate system used here offers several advantages *vis à vis* the speed of a numerical implementation for a given reaction and portability of computer code from reaction to reaction. A

detailed comparison of our results to those of Smith, as well as to related work of other researchers, will be the focus of a separate publication.

As a by-product, this analysis yields an expression for the canonical flux as a function of J . From this a canonical rate coefficient which is a function of J can be derived by substituting Eq. (63) into the J dependent equivalent of Eq. (7),

$$k(T, J) = \frac{g_e}{g_\sigma} \frac{Q_{\text{cons}}(T, R^\dagger)}{Q_{\text{react}}(T)} \frac{2J^2}{\beta^{(n-1)/2}} \left(\frac{2\pi}{h^2} \right)^{(n+1)/2} \\ \times \int d\mathbf{q}_{\text{int}} \left(\frac{|\mathbf{A}_r|}{|\mathbf{B}|} \right)^{1/2} \int_0^\pi d\eta \sin \eta \\ \times \int_0^{2\pi} d\nu \exp(-\beta(V_{\text{eff}}(R^\dagger))), \quad (75)$$

where, again, the optimal value of R for the choice of J and T is $R^\dagger$. This result should be compared with the work of Klotz¹⁶ who has developed expressions for the canonical rate coefficient as function of β and J in which correction factors were introduced that depend on the transition state rotational constant—the present treatment gives a more rigorous definition of this rotational constant. Equation (75), because of the explicit treatment of angular momentum, is anticipated to be more accurate than the standard canonical expression particularly when there is a well defined centrifugal barrier. An investigation of the accuracy and any other computational benefits offered by Eq. (73) will be the subject of future work.

Inspection of Eqs. (73) and (75) shows that $k(T, J)$ is the canonical average of $k(E, J)$ which is as it should be. Inspection also shows that these equations are both dependent on V_{eff} , the effective potential defined in Eq. (74), which includes a term that depends upon the moments of inertia of the overall complex at the transition state. There are, however, no terms relating to coriolis effects despite their presence in the Hamiltonian (see Ref. 10) and the large amplitude motion of the transitional modes. The integration over the internal transitional mode conjugate momenta has effectively incorporated Coriolis effects.

Another benefit of this approach is that the classical Hamiltonian that has been derived here can, via the standard prescription, be used to generate a quantum mechanical Hamiltonian and, hence, used for investigating quantum mechanical effects and corrections. Alternatively, the canonical expressions can be used in conjunction with path integral methods as demonstrated by Klippenstein and Marcus.¹⁷

This work most likely brings to an end one thread of research in FTST begun nearly two decades ago. The inaugural treatment of FTST by Wardlaw and Marcus in the period 1984–1986 determined microcanonical and canonical rate constants by the minimization, along the center-of-mass separation coordinate, of a $2(n+1)$ nested set of phase space integrals over angular momenta and their conjugate angles where n is the number of transitional modes. For the most complicated case of two nonlinear tops ($n=5$), this treatment required the tedious evaluation of a 12-dimensional integral whose value could not be readily physically inter-

preted. Shortly after the introduction of FTST, Klippenstein introduced, developed, and applied the variable reaction coordinate idea, freeing the FTST formalism from a strictly center-of-mass separation requirement. In the remaining years especially over the last decade, Klippenstein, Smith, and the authors have systematically discovered ways, within the context of variable reaction coordinates, of doing many of the phase space integrals analytically, thereby reducing the residual number of integrals that must be done numerically. With this and other recent work by the authors, this residual number of integrals has now been reduced to the minimum possible without specification of the interaction potential. For the energy-resolved microcanonical and for the canonical rate expressions, the dimension of the phase space integral has been reduced from $2(n+1)$ in original FTST to n , with the remaining integrals having the physical interpretation of steric factors for they only involve angular spatial coordinates. For energy- and angular momentum-resolved microcanonical rate expressions, and for the newly recognized angular momentum-resolved canonical rate expression, the dimension has been halved from $2(n+1)$ to $n+1$. The additional integration (beyond n) reduces to an elliptic integral expressible as an analytic series and, under certain circumstances, can be solved in closed form. The reduced dimension of the “new” integrals significantly decreases the numerical effort, allowing quadrature as well as Monte Carlo methods for their evaluation. Future formalistic developments in FTST will most likely concentrate on further reduction in the number of integrals by specification of the form of the interaction potential (e.g., long-range electrostatic interactions) or on higher order generalizations of the variable reaction coordinates and the dividing surfaces they imply.

ACKNOWLEDGMENTS

One of the authors (A.F.W.) was supported by the U.S. Department of Energy, Office of Basic Energy Science, Division of Chemical Sciences, under Contract No. W-31-109-ENG-38. D.M.W. was supported by a research grant from the Natural Sciences and Engineering Research Council of Canada.

APPENDIX A: APPENDICES PROPERTIES OF A AND G MATRICES

Let matrix $\mathbf{A}$ be a real symmetric matrix whose structure can be written as

$$\mathbf{A} = \begin{pmatrix} \mathbf{B} & \mathbf{C} \\ \mathbf{C}^T & \mathbf{D} \end{pmatrix}, \quad (\text{A1})$$

where $\mathbf{B}$ and $\mathbf{D}$ are square symmetric invertible matrices of dimension m and n respectively and $\mathbf{C}$ is a matrix of dimension $m \times n$. Let the inverse matrix of $\mathbf{A}$ be $\mathbf{G}$. $\mathbf{G}$ is also a real symmetric matrix and can be written in a form similar to $\mathbf{A}$,

$$\mathbf{G} = \begin{pmatrix} \mathbf{W} & \mathbf{X} \\ \mathbf{X}^T & \mathbf{Z} \end{pmatrix}, \quad (\text{A2})$$

where $\mathbf{W}$ and $\mathbf{Z}$ are square symmetric invertible matrices of dimension m and n , respectively, and $\mathbf{X}$ is a matrix of dimension $m \times n$.

Since $\mathbf{A}$ and $\mathbf{G}$ are inverse then,

$$\mathbf{A}\mathbf{G} = \mathbf{1}_{m+n}, \quad (\text{A3})$$

where $\mathbf{1}_{m+n}$ is the identity matrix of dimension $m+n$. Substituting Eqs. (A1) and (A2) into Eq. (A3) and performing the multiplication gives,

$$\begin{pmatrix} \mathbf{B}\mathbf{W} + \mathbf{C}\mathbf{X}^T & \mathbf{B}\mathbf{X} + \mathbf{C}\mathbf{Z} \\ \mathbf{C}^T\mathbf{W} + \mathbf{D}\mathbf{X}^T & \mathbf{C}^T\mathbf{X} + \mathbf{D}\mathbf{Z} \end{pmatrix} = \mathbf{1}_{m+n}. \quad (\text{A4})$$

Equating blocks gives

$$\mathbf{B}\mathbf{W} + \mathbf{C}\mathbf{X}^T = \mathbf{1}_m, \quad (\text{A5})$$

$$\mathbf{B}\mathbf{X} + \mathbf{C}\mathbf{Z} = \mathbf{0}_{m \times n}, \quad (\text{A6})$$

$$\mathbf{C}^T\mathbf{W} + \mathbf{D}\mathbf{X}^T = \mathbf{0}_{n \times m}^T, \quad (\text{A7})$$

$$\mathbf{C}^T\mathbf{X} + \mathbf{D}\mathbf{Z} = \mathbf{1}_n. \quad (\text{A8})$$

Two useful relations can be derived from Eqs. (A5)–(A8) as follows: Equation (A6) implies,

$$\mathbf{C} = -\mathbf{B}\mathbf{X}\mathbf{Z}^{-1}. \quad (\text{A9})$$

Substitution in Eq. (A5) followed by rearrangement gives

$$\mathbf{B}^{-1} = \mathbf{W} - \mathbf{X}\mathbf{Z}^{-1}\mathbf{X}^T. \quad (\text{A10})$$

Similarly Eq. (A6) implies,

$$\mathbf{X} = -\mathbf{B}^{-1}\mathbf{C}\mathbf{Z}. \quad (\text{A11})$$

Substitution in Eq. (A8) followed by rearrangement gives

$$\mathbf{Z}^{-1} = \mathbf{D} - \mathbf{C}^T\mathbf{B}^{-1}\mathbf{C}. \quad (\text{A12})$$

Equation (A10) is used in Sec. III.

APPENDIX B: IMPLEMENTATION ISSUES

In this Appendix the remaining integrals in Eq. (72) are addressed. The transitional mode potential is determined by the system in question and so it is not possible to derive a general solution for every system. Indeed, with the exception of some particularly simple potentials this integral must be tackled using numerical methods. However, the integral over η can be evaluated analytically if the following substitutions are made:

$$\epsilon = E - V_{\text{tr}}(\mathbf{q}_{\text{int}}), \quad (\text{B1})$$

$$x = \cos \eta. \quad (\text{B2})$$

If $\mathcal{I}$ denotes the integral over η and ν , Eq. (72) becomes

$$\begin{aligned} f(E, J, R) &= \frac{2J^2}{\Gamma((n-1)/2)} \left(\frac{2\pi}{h^2} \right)^{(n+1)/2} \\ &\times \int d\mathbf{q}_{\text{int}} \left(\frac{|\mathbf{A}_r|}{|\mathbf{B}|} \right)^{1/2} \mathcal{I}, \end{aligned} \quad (\text{B3})$$

and, with the above substitutions, $\mathcal{I}$ is given by

$$\mathcal{I} = \int_0^{2\pi} d\nu \int_{-1}^1 dx (\epsilon - E_{\text{rot}}(J))^{(n-3)/2} u(\epsilon - E_{\text{rot}}(J)), \quad (\text{B4})$$

where

$$E_{\text{rot}}(J) = J^2 [(\sin^2 \nu / I_a + \cos^2 \nu / I_b)(1 - x^2) + x^2 / I_c] / 2. \quad (\text{B5})$$

In order to simplify the notation it is useful to introduce the term κ defined as,

$$\kappa = I_c(\sin^2 \nu / I_a + \cos^2 \nu / I_b), \quad (\text{B6})$$

so that $E_{\text{rot}}(J)$ in terms of κ is given by

$$E_{\text{rot}}(J) = \frac{J^2}{2I_c} (\kappa + (1 - \kappa)x^2). \quad (\text{B7})$$

The step function in Eq. (72) constrains the values that x can have, and therefore determines the range of integration. To find the limits on x it is necessary to determine the roots of the step function condition, i.e., to solve

$$\epsilon - \frac{J^2}{2I_c} (\kappa + (1 - \kappa)x^2) = 0 \quad (\text{B8})$$

for x . The roots can be written as

$$x_{\pm} = \pm P/Q, \quad (\text{B9})$$

where P and Q are defined by

$$P^2 = \epsilon - \frac{J^2}{2I_c} \kappa, \quad (\text{B10})$$

$$Q^2 = \frac{J^2}{2I_c} (1 - \kappa). \quad (\text{B11})$$

It is of interest to examine these roots. If the moments of inertia of a given configuration are ordered such that $I_a \geq I_b \geq I_c > 0$, then $0 < \kappa \leq 1$, and in fact it can only be unity for two cases: a spherical top situation which cannot occur for a transition state and the case where $I_b = I_c$ and $\cos^2 \nu = 1$. This latter case corresponds to an oblate symmetric top with the angular momentum perpendicular to the figure axis, again a configuration that no real transition state is likely to adopt. It follows, therefore that Q^2 is always positive, and Q is always real. On the other hand, P^2 can be negative since ϵ can be strongly negative for those parts of phase space that are classically forbidden and, even for small positive values of ϵ , P^2 can still be negative due to centrifugal barriers. Thus for a numerical procedure the value of P^2 will determine whether a given configuration will make a contribution, configurations with $P^2 < 0$ being rejected.

The roots $x_{\pm}$ indicate that there are three possible situations that can occur: first of these is that the roots are imaginary and the integral, and therefore, the contribution to $f(E, J, R)$, is zero. The second situation is when $0 < |x_{\pm}| < 1$. In this case $\mathcal{I}$ can be written as

$$\mathcal{I} = 2 \int_0^{2\pi} d\nu \int_0^{P/Q} dx (P^2 - Q^2 x^2)^{(n-3)/2}, \quad (\text{B12})$$

where the fact that the integrand is even in x has been used to reduce the range of integration. It is straightforward to show that this integral can be reduced to the form,

$$\mathcal{I} = 2 \int_0^{2\pi} d\nu \frac{P^{n-2}}{Q} \int_0^1 dy (1 - y^2)^{(n-3)/2}, \quad (\text{B13})$$

where $y = (Q/P)x$. The last integral is of standard form and gives

$$\mathcal{I} = 2 \int_0^{2\pi} d\nu \frac{P^{n-2}}{Q} \left[\frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \cdots \left(\frac{\pi}{2} \right)^l \right], \quad (\text{B14})$$

where l is unity if n is even and zero if n is odd.

The third situation is where $x_+ > 1$ and since x cannot be greater than one for there to be a meaningful value of η , then the equivalent of Eq. (B12) is

$$\mathcal{I} = 2 \int_0^{2\pi} d\nu \int_0^1 dx (P^2 - Q^2 x^2)^{(n-3)/2} \quad (\text{B15})$$

$$= 2 \int_0^{2\pi} d\nu \frac{P^{n-2}}{Q} \int_0^{Q/P} dy (1 - y^2)^{(n-3)/2}. \quad (\text{B16})$$

The integral over y can be evaluated by recursion and it can be shown that

$$\begin{aligned} \mathcal{I} = 2 \int_0^{2\pi} d\nu \frac{P^{n-2}}{Q} & \left[\frac{\alpha(1-\alpha^2)^{(n-3)/2}}{(n-2)} \right. \\ & + \frac{(n-3)}{(n-2)} \frac{\alpha(1-\alpha^2)^{(n-5)/2}}{(n-4)} + \cdots \\ & \left. + \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \cdots h(\alpha) \right], \end{aligned} \quad (\text{B17})$$

where $\alpha = Q/P$ and the form of $h(\alpha)$ depends on the value of n ,

$$h(\alpha) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ \arcsin \alpha & \text{if } n \text{ is even.} \end{cases} \quad (\text{B18})$$

Note Eq. (B14) is recovered if $\alpha = 1$.

The integral $\mathcal{I}$ has now been reduced to two equations, Eqs. (B14) and (B17), that are each to be evaluated for $n = 4, 5, 6, 7$, and 8 . This gives ten cases altogether. In each case, there remains an integral over ν . As will be shown in the rest of this Appendix, in two of the ten cases, this integration is analytic while in the remaining cases, the integration can be expressed in terms of elliptic integrals. Elliptic integrals are susceptible to exact numerical evaluation, table lookup, or polynomial approximation. However, the expressions developed in this Appendix that incorporate elliptic integrals do grow in complexity for the larger values of n . It remains to be seen in actual applications if these expressions are more efficient than direct evaluation of Eqs. (B14) and (B17) by standard numerical methods.

To proceed with the integration over ν , the properties of κ must first be examined in more detail. Because of the ordering of $I_a \geq I_b \geq I_c$, κ can be rewritten as

$$\kappa = \kappa_L \sin^2 \nu + \kappa_H \cos^2 \nu = \kappa_L + \Delta \cos^2 \nu, \quad (\text{B19})$$

where κ_L and κ_H are the lowest and highest values that κ can attain and $\Delta = \kappa_H - \kappa_L$. By construction $1 \geq \kappa_H > \kappa_L > 0$ and the extreme values are reached at $\nu = 0$ and $\pi/2$ and symmetrically related values in the rest of the 0 to 2π range of ν . There are two consequences of Eq. (B19). First, the symmetry implicit in Eq. (B19) means that the integration over ν in Eqs. (B14) and (B17) need only be done from 0 to $\pi/2$ and

the results multiplied by four because all ν dependence in these two equations is exclusively through κ . Second, the solution $x_{\pm}$ can be compactly rewritten,

$$x_{\pm} = [(\rho - \kappa)/(1 - \kappa)]^{1/2}, \quad (\text{B20})$$

where $\rho = \epsilon/(J^2/2I_c)$. If $\rho > 1$, then from Eq. (B20), $|x_{\pm}| > 1$, regardless of the value of κ . Since $\kappa_H < 1$, over the full range of ν integration, $|x_{\pm}|$ always remains greater than 1 and Eq. (B17) remains as written with no restriction on the ν integration. However, if $\rho < 1$, then $|x_{\pm}|$ can either be imaginary or real and < 1 depending on the value of κ or, in other words, on the value of ν . Since imaginary $x_{\pm}$ leads to no contribution to $\mathcal{I}$, this means that there are limits on the range of integration over ν that have not been specified in Eq. (B14). Rewriting Eq. (B14) with the limits explicit gives

$$\begin{aligned} \mathcal{I} = 8 & \left[\frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \cdots \left(\frac{\pi}{2} \right)^l \right] \left(\frac{J^2}{2I_c} \right)^{n-3} \\ & \times \int_{\nu_{mn}}^{\pi/2} \frac{(\rho - \kappa)^{(n-2)/2}}{(1 - \kappa)^{1/2}} d\nu \\ = C'_n & \int_{\nu_{mn}}^{\pi/2} \frac{(\rho - \kappa)^{(n-2)/2}}{(1 - \kappa)^{1/2}} d\nu, \end{aligned} \quad (\text{B21})$$

where

$$\nu_{mn} = \begin{cases} 0 & \text{if } \rho \geq \kappa_H \\ \arccos[(\rho - \kappa_L)/\Delta]^{1/2} & \text{if } \kappa > \rho > \kappa_L \\ \pi/2 & \text{if } \kappa_L \geq \rho, \end{cases} \quad (\text{B22})$$

and where the κ dependence of the integrand has been made explicit. For convenience in Eq. (B21), the complicated prefactor has been gathered into the constant C'_n . Note from Eq. (B22), if $\rho \leq \kappa_L$, there is no allowable range of ν , implying $f(E, J, R) = 0$.

With the limitations on the range of integration over ν in place, each integral can now be performed for each value of n . We will do this for Eq. (B21) first, and then Eq. (B17).

For Eq. (B21) under the condition that $\rho > \kappa_L$, substitution of Eq. (B19) for κ followed by the change of variables of $u = \cos \nu$ and a gathering of constants gives

$$\begin{aligned} \mathcal{I} = C'_n & \frac{(\rho - \kappa_L)^{(n-2)/2}}{(1 - \kappa_L)^{1/2}} \\ & \times \int_0^{u_{mx}} \frac{(1 - Au^2)^{(n-2)/2}}{[(1 - u^2)(1 - Bu^2)]^{1/2}} du \\ = C_n & \int_0^{u_{mx}} \frac{(1 - Au^2)^{(n-2)/2}}{[(1 - u^2)(1 - Bu^2)]^{1/2}} du, \end{aligned} \quad (\text{B23})$$

where

$$\begin{aligned} A &= \Delta/(\rho - \kappa_L), \\ B &= \Delta/(1 - \kappa_L), \end{aligned} \quad (\text{B24})$$

$$u_{mx} = [(\kappa_m - \kappa_L)/\Delta]^{1/2},$$

and where $\kappa_m = \min(\rho, \kappa_H)$ and all the constants in the prefactor have been gathered into the constant C_n . When n is even, the integrand is a fraction whose denominator is the

square root of a quartic polynomial and whose numerator is a polynomial. Such integrals can be expressed in terms of elliptic integrals,

$$\mathcal{I} = C_n \sum_{i=0}^m \frac{m!(-A)^i}{i!(m-i)!} L_{2i}, \quad (\text{B25})$$

where

$$\begin{aligned} L_0 &= F(\phi, B), \\ L_2 &= \frac{1}{B} [F(\phi, B) - E(\phi, B)], \\ L_s &= \frac{1}{B(s-1)} [u_{mx}^{s-3} y_{mx} + (1+B)(s-2)L_{s-2} \\ &\quad - (s-3)L_{s-4}] \quad s=4, 6, \dots, \end{aligned} \quad (\text{B26})$$

and where $m=(n-2)/2$, $\phi = \arcsin(u_{mx})$, F and E are Legendre elliptic integrals of the first and second kind, and $y_{mx} = [(1-u_{mx}^2)(1-Bu_{mx}^2)]^{1/2}$.

When n is odd, i.e., $n=2m+1$, Eq. (B23) can be transformed as follows:

$$\begin{aligned} \mathcal{I} &= C_n \int_0^{u_{mx}} \frac{(1-Au^2)^m}{[(1-Au^2)(1-u^2)(1-Bu^2)]^{1/2}} du \\ &= (C_n/2) \int_0^{u_{mx}} \frac{(1-Au^2)^m}{[u^2(1-Au^2)(1-u^2)(1-Bu^2)]^{1/2}} 2udu \\ &= (C_n/2) \int_0^{z_{mx}} \frac{(1-Az)^m}{[z(1-Az)(1-z)(1-Bz)]^{1/2}} dz \\ &= (C_n/2) \frac{(\xi\xi)^{1/2}}{\xi^m} \int_0^1 \frac{(\xi-x)^m}{[x(\xi-x)(\gamma-x)(\xi-x)]^{1/2}} dx, \end{aligned} \quad (\text{B27})$$

where

$$\begin{aligned} \xi &= (\rho - \kappa_L)/(\kappa_m - \kappa_L) = 1/(Au_{mx}^2), \\ \zeta &= (1 - \kappa_L)/(\kappa_m - \kappa_L) = 1/(Bu_{mx}^2), \\ \gamma &= (\kappa_H - \kappa_L)/(\kappa_m - \kappa_L) = 1/u_{mx}^2. \end{aligned} \quad (\text{B28})$$

Once again the form of the integrand is that of a fraction with a denominator that is the square root of a quartic polynomial and with a numerator that is a polynomial. Such integrals can be expressed in terms of elliptic integrals,

$$\mathcal{I} = C_n \frac{1}{\xi^m} \left[\frac{\xi\zeta}{(\zeta-1)\psi} \right]^{1/2} \sum_{i=0}^m \frac{(\xi-\zeta)^i m!}{i!(m-i)!} L'_i, \quad (\text{B29})$$

where

$$\begin{aligned} L'_0 &= F(\pi/2, r), \\ L'_1 &= \zeta \Pi(\pi/2, 1/(\zeta-1), r), \\ L'_2 &= \frac{1}{2} \left\{ (3\zeta - \xi - \gamma) L'_1 - \frac{(\zeta - \xi)(\zeta - \gamma)}{(\zeta - \psi)} \right. \\ &\quad \left. \times [\zeta F(\pi/2, r) - \psi E(\pi/2, r)] \right\}, \end{aligned}$$

$$\begin{aligned} L'_3 &= \frac{3(3\zeta - \xi - \gamma)}{4} L'_2 - \frac{[\xi\gamma + \zeta(3\zeta - 2\xi - 2\gamma)]}{2} L'_1 \\ &\quad + \frac{\zeta(\zeta - \xi)(\zeta - \gamma)}{4} L'_0, \end{aligned} \quad (\text{B30})$$

and where Π is the Legendre elliptic integrals of the third kind, $r = [(\zeta - \psi)/(\zeta - 1)/\psi]^{1/2}$, and $\psi = \max(\zeta, \gamma)$.

Elliptic integrals do not have consistent notation in the literature. The notation used in Eqs. (B26) and (B27) is that found in Ref. 18, where efficient algorithms can be found for all the elliptic integrals appearing in the equations. If $\phi = \pi/2$, the elliptic integrals are said to be complete. Elliptic integrals Π of the third kind only appear in the above equations in complete form. Elliptic integrals F and E are both complete and incomplete in the above equations.

Concerning Eq. (B17), the bracket inside the integrand involves two kinds of terms. For all n , one kind of term involves the product of powers of α times square roots of $1 - \alpha^2$. For n even, there is second kind of term involving $\arcsin \alpha$. The part of the integral over ν in Eq. (B17) that involves the first kind of terms is analytic. Consequently, Eq. (B17) for $n=4-8$ takes the form

$$\mathcal{I}_4 = 2\pi(\epsilon - c_J)^{1/2} + 4\mathcal{I}'_4, \quad (\text{B31})$$

$$\mathcal{I}_5 = 4\pi[\epsilon - \frac{1}{3}(1 + \kappa_H + \kappa_L)c_J], \quad (\text{B32})$$

$$\mathcal{I}_6 = \frac{\pi}{4}(\epsilon - c_J)^{1/2}[10\epsilon - (4 + 3\kappa_H + 3\kappa_L)c_J] + 3\mathcal{I}'_6, \quad (\text{B33})$$

$$\begin{aligned} \mathcal{I}_7 &= \frac{4\pi}{15}[15\epsilon^2 - 10\epsilon c_J + 3c_J^2 - 4\epsilon(\epsilon + c_J)(\kappa_H + \kappa_L) \\ &\quad + (8\kappa_H\kappa_L + 3\Delta^2)c_J^2], \end{aligned} \quad (\text{B34})$$

$$\begin{aligned} \mathcal{I}_8 &= \frac{\pi}{12}(\epsilon - c_J)^{1/2}[33\epsilon^2 - 26\epsilon c_J + 8c_J^2 - 5c_J(4\epsilon + c_J) \\ &\quad \times (\kappa_H + \kappa_L) + 15(\kappa_H\kappa_L - 3\Delta^2/8)c_J^2] + \frac{5}{2}\mathcal{I}'_8, \end{aligned} \quad (\text{B35})$$

where $c_J = J^2/(2I_C)$. Note that for $n=5$ and 7, the integral is directly expressible in terms of elementary functions.

For $n=4, 6$, and 8 in Eqs. (B31), (B33), and (B35), there remains a residual integral $\mathcal{I}'_n$ of the type,

$$\mathcal{I}'_n = c_J^{(n-3)/2} \int_0^{\pi/2} (\rho - \kappa)^{(n-3)/2} \left[\frac{\arcsin \alpha}{\alpha} \right] d\alpha, \quad (\text{B36})$$

where $\alpha = [(1 - \kappa)/(\rho - \kappa)]^{1/2}$. This integral can not apparently be directly reduced to expressions involving elementary and/or special functions. However, $(\arcsin \alpha)/\alpha$ can be expanded about $\alpha=0$ in terms of powers of α^2 which reduces $\mathcal{I}'_n$ into an infinite series of integrals, each term of which, to within a scale factor, has the form

$$\begin{aligned}
\text{term}(l) &= \int_0^{\pi/2} (\rho - \kappa)^{(n-3)/2} [(1 - \kappa)/(\rho - \kappa)]^l d\nu \\
&= \int_0^{\pi/2} (\rho - \kappa)^{(n-2l-3)/2} (1 - \kappa)^l d\nu \\
&= (\rho - \kappa_L)^{(n-3)/2} \alpha_L^{2l} \\
&\quad \times \int_0^1 \frac{(1 - Au^2)^{(n-2l-2)/2} (1 - Bu^2)^l}{[(1 - u^2)(1 - Au^2)]^{1/2}} du,
\end{aligned}
\tag{B37}$$

where $\alpha_L = [(1 - \kappa_L)/(\rho - \kappa_L)]^{1/2}$ and is less than zero. Since $(n - 2l - 2)/2$ is an integer, the properties of elliptic integrals allow Eq. (B37) to be relatively directly expressed as a linear combination of all three complete elliptic integrals where the combination coefficients are algebraic expressions. Then ultimately Eq. (B36) can be expressed as a linear combination of the three complete elliptic functions where each combination coefficient is an infinite series in α_L^2 with its own coefficients which are complicated algebraic expressions. Since such a solution would be slowly convergent as $\alpha_L^2 \rightarrow 1$, this is not a useful way to rigorously evaluate Eq. (B36). However, if $(\arcsin \alpha)/\alpha$ is fit to finite expansion in α^2 where the fitting coefficients minimize error over the whole range of α from 0 to 1, then this may be a relatively rapid way to approximate Eq. (B36). For example an optimized quadratic expansion in

α of $(\arcsin \alpha)/\alpha$ has a RMS error of 1.5% over the range of $[0, 1]$ and a maximum error of 10.0%. A optimized linear expansion has a RMS error of 2.9% and a maximum error of 15.9%. Thus an approximation to Eq. (B36) good to $\approx 10\%$ or better would require the explicit evaluation of only two or three terms of the form of Eq. (B37).

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